CS & IT ENGINEERING

Operating System

Deadlock

Lecture - 3

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Recap of Previous Lecture

doubt ?

Topic Deadlock

Topic Deadlock Prevention

Topics to be Covered

Topic Deadlock Detection

Topic Banker's Algorithm

#Q.

Safe

Process	Allocation A B C D	Max A B C D	Available A B C D
P1	0 0 1 2	0 0 1 2	1 5 2 0
P2	1 0 0 0	1 7 5 0	
P3	1 3 5 4	2 3 5 6	
P4	0 6 3 2	0 6 5 4	
P5	0 0 1 4	0 6 5 6	

Topic : Banker's Algorithm

Process	Allocation	Max	Available	Need
	A B C	A B C	A B C	A B C
P_0	0 1 0	7 5 3	~~3 3 2~~ ~~2 3 0~~ 2 2 0	7 4 3
P_1	~~2 0 0~~ 3 0 2	3 2 2		~~1 2 2~~ 0 2 0
P_2	3 0 2	9 0 2		6 0 0
P_3	~~2 1 1~~ 2 2 1	2 2 2		~~0 1 1~~ 0 0 1
P_4	0 0 2	4 3 3		4 3 1

[NAT]

granted

#Q. What will happen if process P1 requests one additional instance of resource type A and two instances of resource type C ? Request$_{P_1}$ = $\langle 1, 0, 2 \rangle$

What will happen if process P3 requests one additional instance of resource type B?

Request P_3 = $\langle 0, 1, 0 \rangle$

Both requests are granted.

Cases :-

1. Both requests are granted
2. Both requests are rejected
3. only Req1 granted but req2 rejected
4. only Req2 granted but req1 rejected
5. Either Req1 granted or req2 granted but not both together.

Topic : Deadlock Detection

1. When all resources have single instance
2. When resources have multiple instances

Topic : Deadlock Detection

When all resources have single instance:

Deadlock detection is done using wait-for-graph

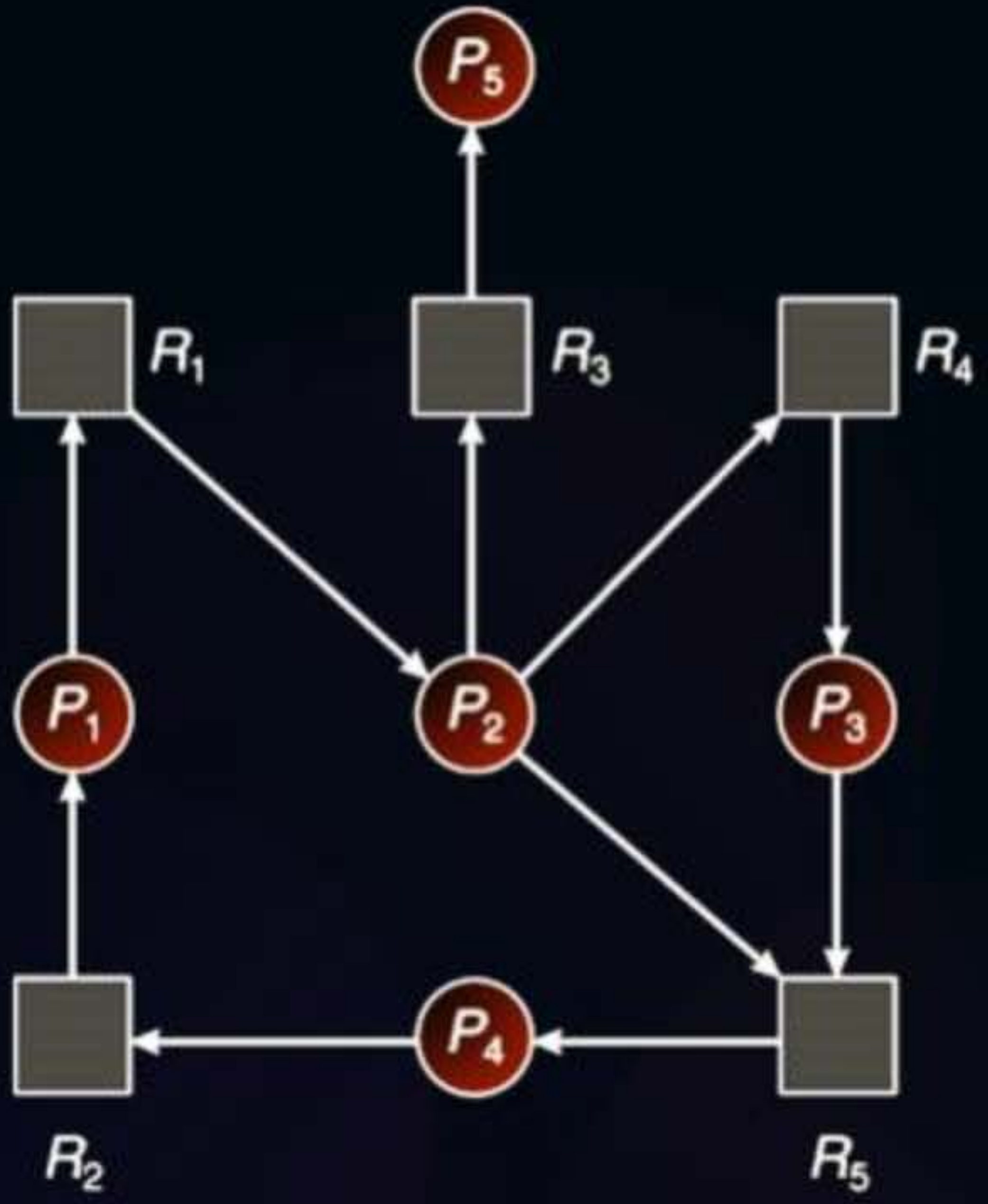
Topic : Wait For Graph

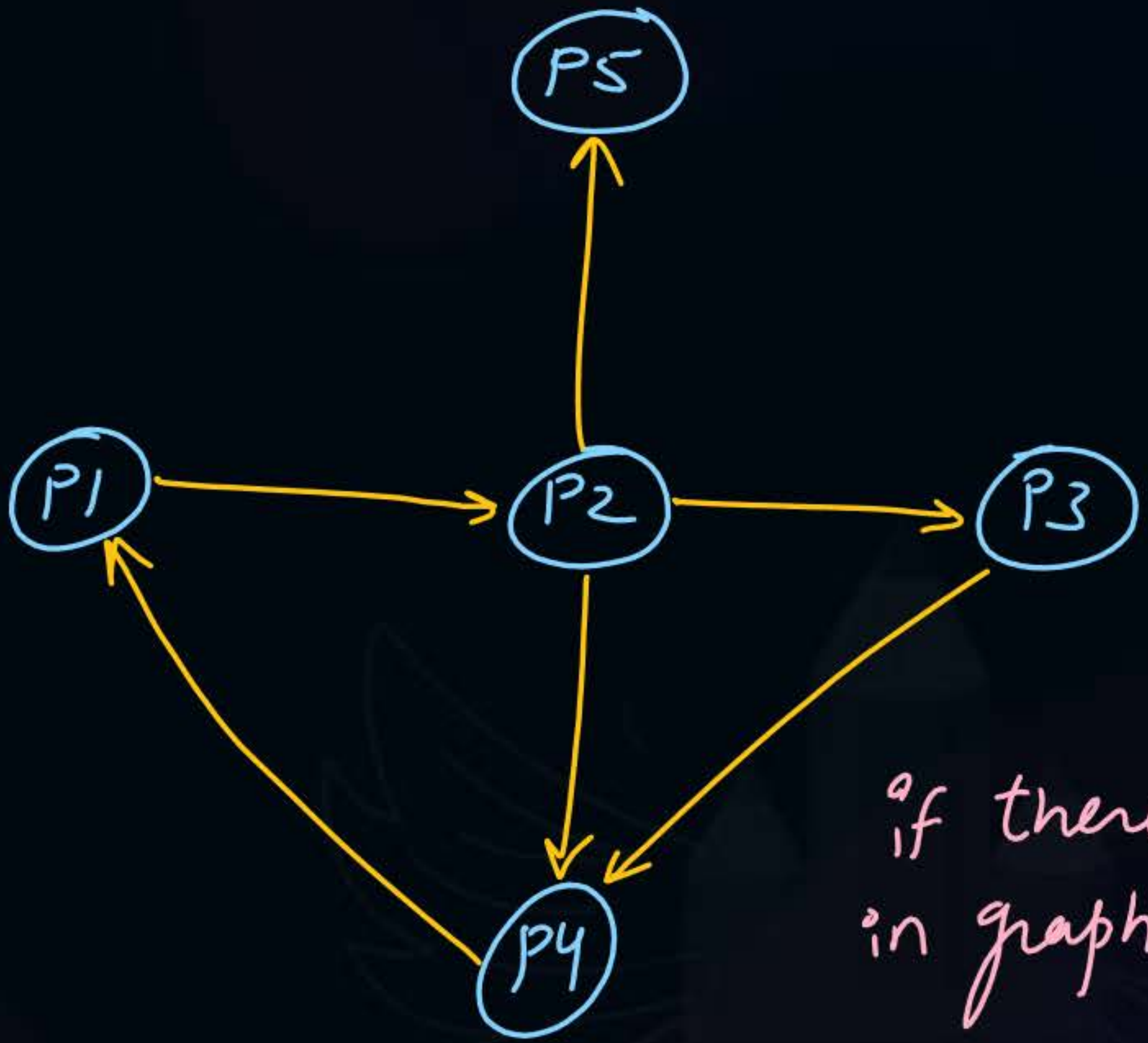
It is created from resource allocation graph $\Rightarrow$ which includes only processes.

edge from $P_i \longrightarrow P_j$

means process P_i is waiting-for a resource which is with P_j.

Topic : Wait For Graph





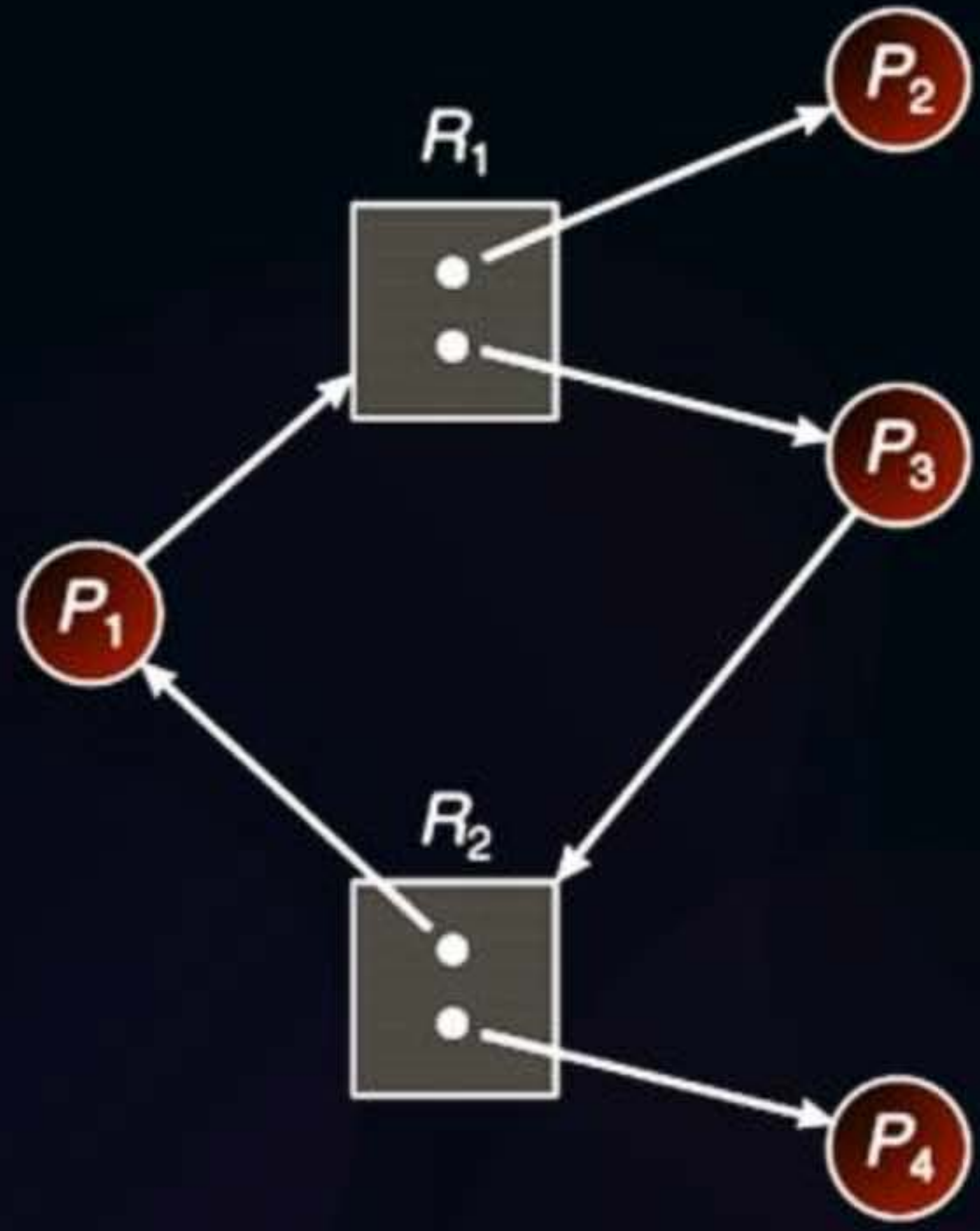
if there is a cycle in graph then deadlock.

Topic : Wait For Graph

If a resource category contains more than one instance, then the presence of a cycle in the resource-allocation graph indicates the possibility of a deadlock, but does not guarantee one.

wait-for graph

Topic : Wait For Graph: Example



wait-for graph

cycle but no deadlock

	Allocation		Request		Available	
	R1	R2	R1	R2	R1	R2
P1	0	1	1	0	0	0
P2	1	0	0	0		
P3	1	0	0	1		
P4	0	1	0	0		

Topic : Deadlock Detection

When resources have multiple instance:

Deadlock detection is done using a specific algorithm

→ banker's algo

Topic : Deadlock Detection Algorithm

	Allocation	Request	Available
	A B C	A B C	A B C
P_0	0 1 0	0 0 0	~~0 0 0~~
P_1	2 0 0	2 0 2	After P_0 ⇒ 0 1 0
P_2	3 0 3	0 0 0	After P_2 ⇒ 3 1 3
P_3	2 1 1	1 0 0	After P_1 ⇒ 5 1 3
P_4	0 0 2	0 0 2	After P_3 ⇒ 7 2 4

After P_4 ⇒ 7 2 6

All processes can run
⇓
no deadlock

Ques)

	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
P_0	1	0	1	0	0	1	0	0	1
P_1	1	2	1	1	1	3			
P_2	2	1	1	2	1	3			
P_3	2	2	1	1	2	4			

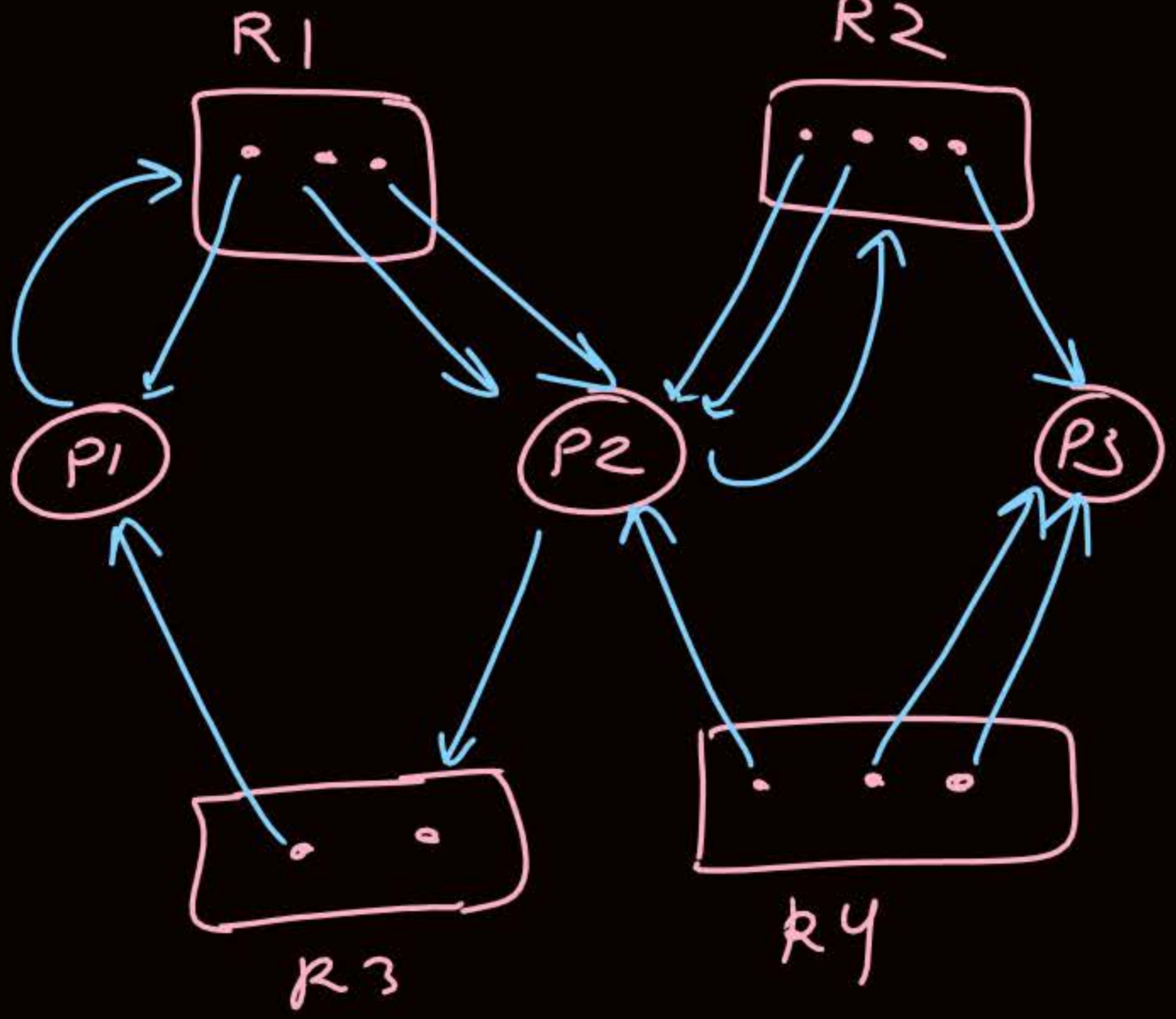
after P_0 1 0 2

$\Downarrow$

$\boxed{\text{deadlock}}$

$\Downarrow$

P_1, P_2, P_3 are deadlocked



	Allocation				Request			
	R1	R2	R3	R4	R1	R2	R3	R4
P1	1	0	1	0	1	0	0	0
P2	2	2	0	1	0	1	1	0
P3	0	1	0	2	0	0	0	0

Available

	R1	R2	R3	R4
	0	1	1	0
after P2	2	3	1	1

→ all processes can run

⇑

no deadlock

Topic : Detection-Algorithm Usage

When should the deadlock detection be done? Frequently, or infrequently?

detection algo runs
when OS allocates a
resource to a process.

Topic : Recovery From Deadlock

There are three basic approaches to recovery from deadlock:

1. Inform the system operator and allow him/her to take manual intervention
2. Terminate one or more processes involved in the deadlock
3. Preempt resources.

Topic : Process Termination

Terminate all processes involved in the deadlock

Terminate processes one by one until the deadlock is broken

Topic : Resource Preemption

Important issues to be addressed when preempting resources to relieve deadlock:

1. Selecting a victim process
2. Rollback
3. Starvation

ex:-

	max	Allocation
A	2	1
B	3	1
C	2	1

deadlock

3 instances

Allocation
1
2
1

deadlock

4 instances

Allocation
1
2
1

~~5~~ instances

1

$\Downarrow$

available

[NAT]

#Q. Consider a system with 3 processes A, B and C. All 3 processes require 4 resources each to execute. The minimum number of resources the system should have such that deadlock can never occur?

	max	Allocation	available
A	4	3	1
B	4	3	
C	4	3	

$3+3+3+1 = 10$

[NAT]

#Q. Consider a system with 4 processes A, B, C and D. All 4 processes require 6 resources each to execute. The maximum number of resources the system should have such that deadlock may occur?

	max	allocation
A	6	5
B	6	5
C	6	5
D	6	5
		20 Ans.

if n processes

each requires k instances

for no deadlock $\Rightarrow$

$$n(k-1) + 1 \leq \text{total resources}$$

[NAT]

#Q. Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is 2.

$$3(k-1)+1 \leq 4$$

$$3k-3 \leq 3$$

$$3k \leq 6$$

$$k \leq 2$$

$$\boxed{k_{max} = 2}$$

Ques) n processes
each requires 4 instances
Total instances = 24
max value of n for which no deadlock ?

Sol^n

$$n(4-1)+1 \le 24$$

$$3n \le 23$$

$$n \le 7.66$$

$$\boxed{n_{max} = 7}$$

Topic : Types of Locks

1. ~~1.~~ Spinlock → busy waiting
2. Livelock → processes keeps moving b/w running and ready state.
3. Deadlock → all the deadlocked processes will be in blocked state.

at least 2 process

4. Semaphores
5. Reentrant Locks → one process can lock an item multiple time without releasing.

ex:- binary semaphore

$S1 = \cancel{1}\ 0$

$S2 = \cancel{1}\ 0$

P1	P2
wait(S1)	wait(S2)
→ wait(S2)	→ wait(S1)
≡	≡
signal(S1)	signal(S2)
signal(S2)	signal(S1)

P1 and P2 are in $\boxed{\text{livelock}}$

⇓

deadlock situation

2 mins Summary

Topic Deadlock Detection

Topic Banker's Algorithm

Happy Learning

THANK - YOU